

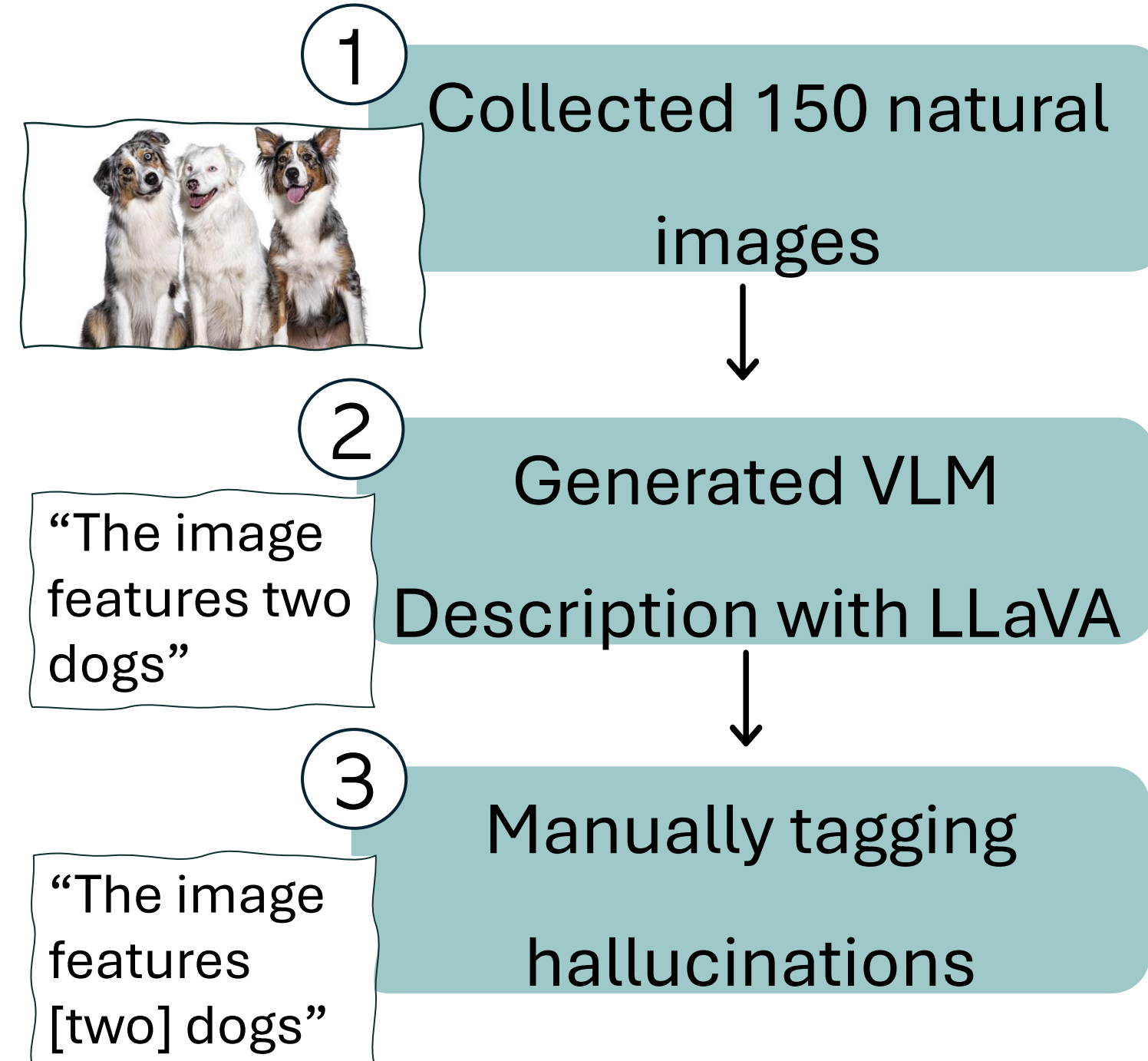
Hallucinations in Vision-Language Models (VLM)

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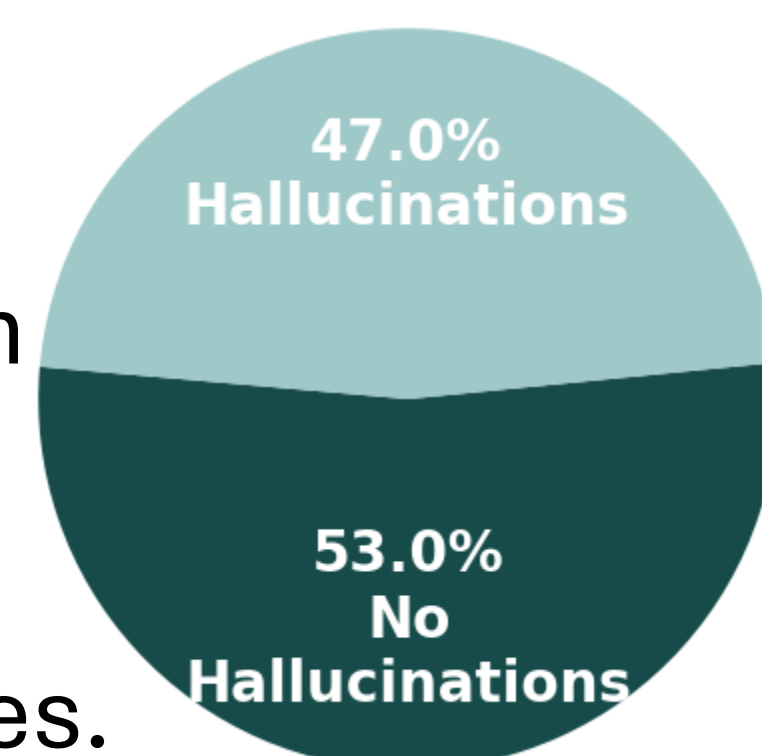
Data Collection

Dataset Generation



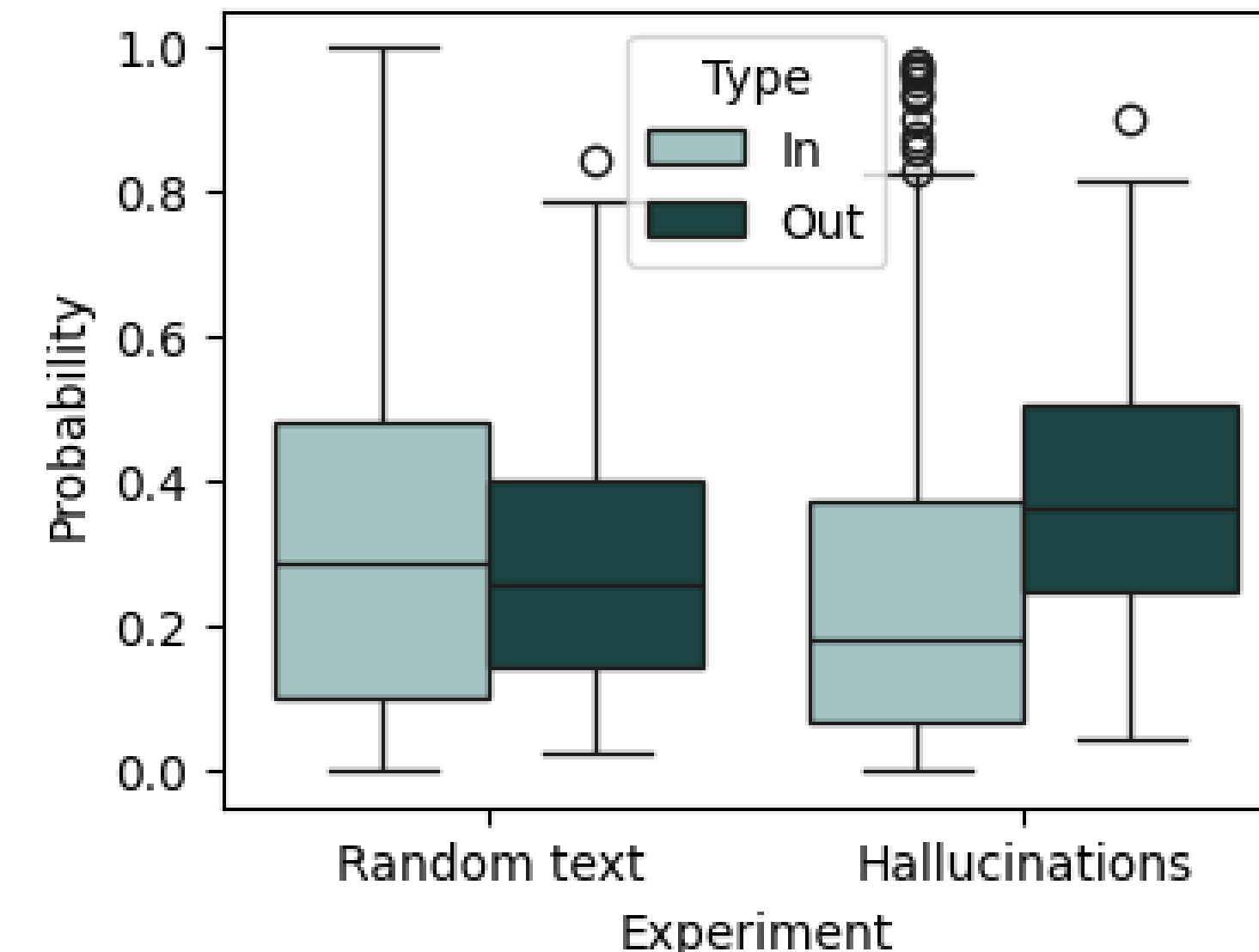
Pre-Processing

- Data Validation (brackets), cleaning, stop words removal, ...
- Mapping between token's logits to words probabilities.
- Splitting the description into sentences dataset with total of 761 samples.

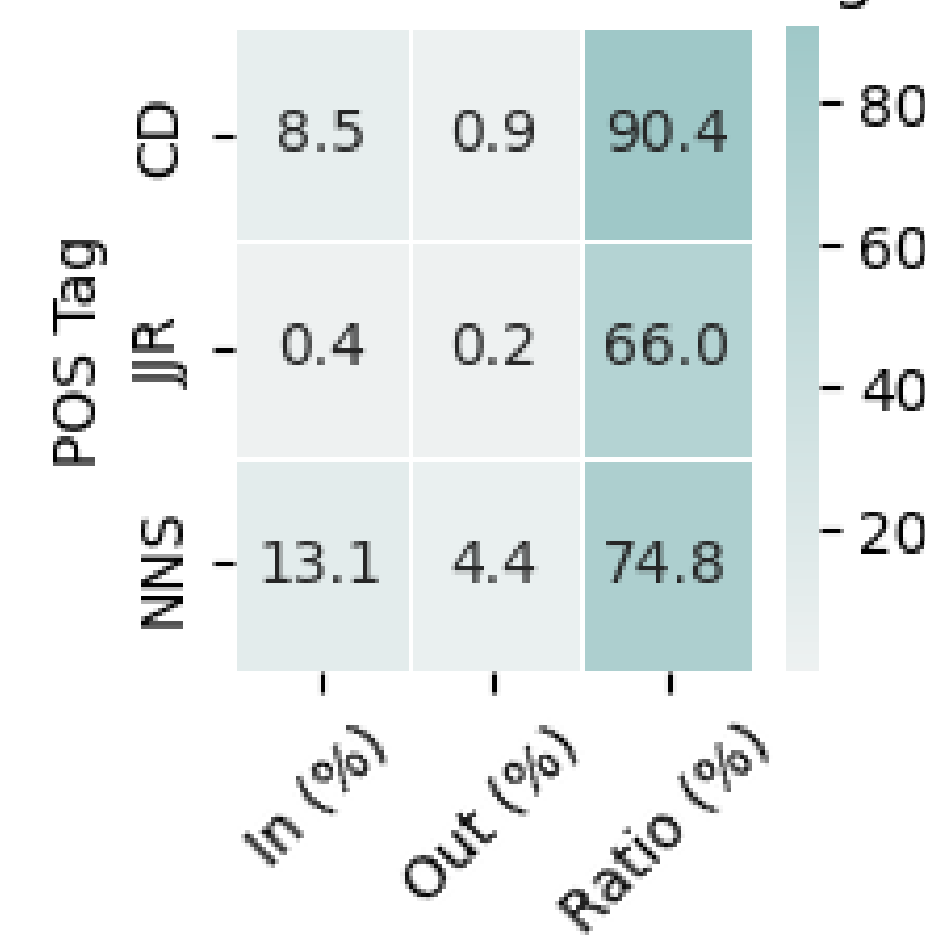


EDA

median probabilities inside and outside hallucinations



Distribution of POS Tags



Feature Extraction

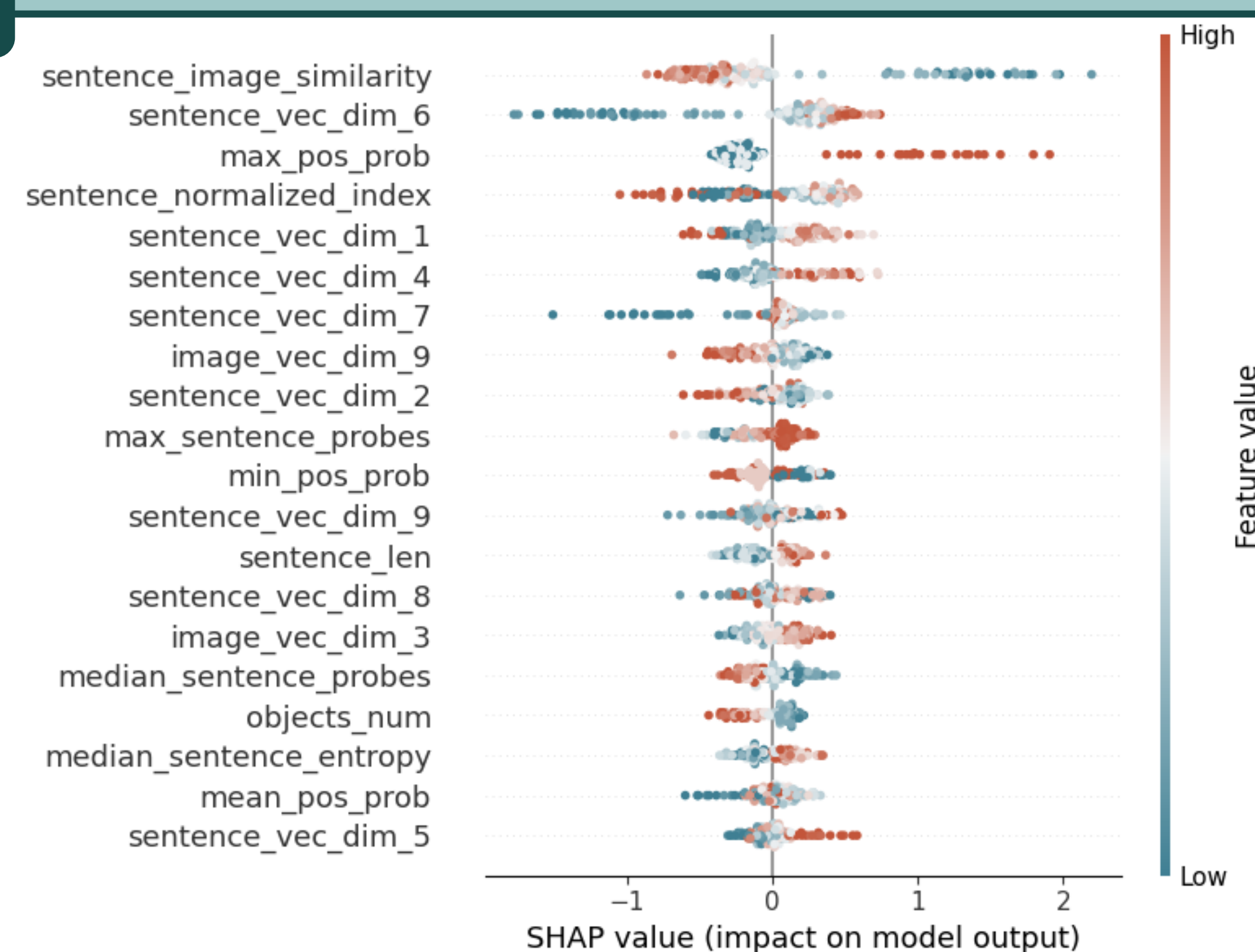
POS Contains JJR/NNS/CD, min/mean/max POS probability for hallucination

Image Number of objects (YOLO), mean R/G/B

Sentence Length, normalized index in the description, is contains hedges

CLIP Extracted image and sentence embedding → Calculate cosine similarity → Reduced to 10 dimensions by PCA

Logits Probability/Entropy aggregations (min, max, mean, median)



Methods

Heuristic

Predicting hallucinations based on logits and POS only:

Algorithm

Input: sentence, words_logit, words_POS

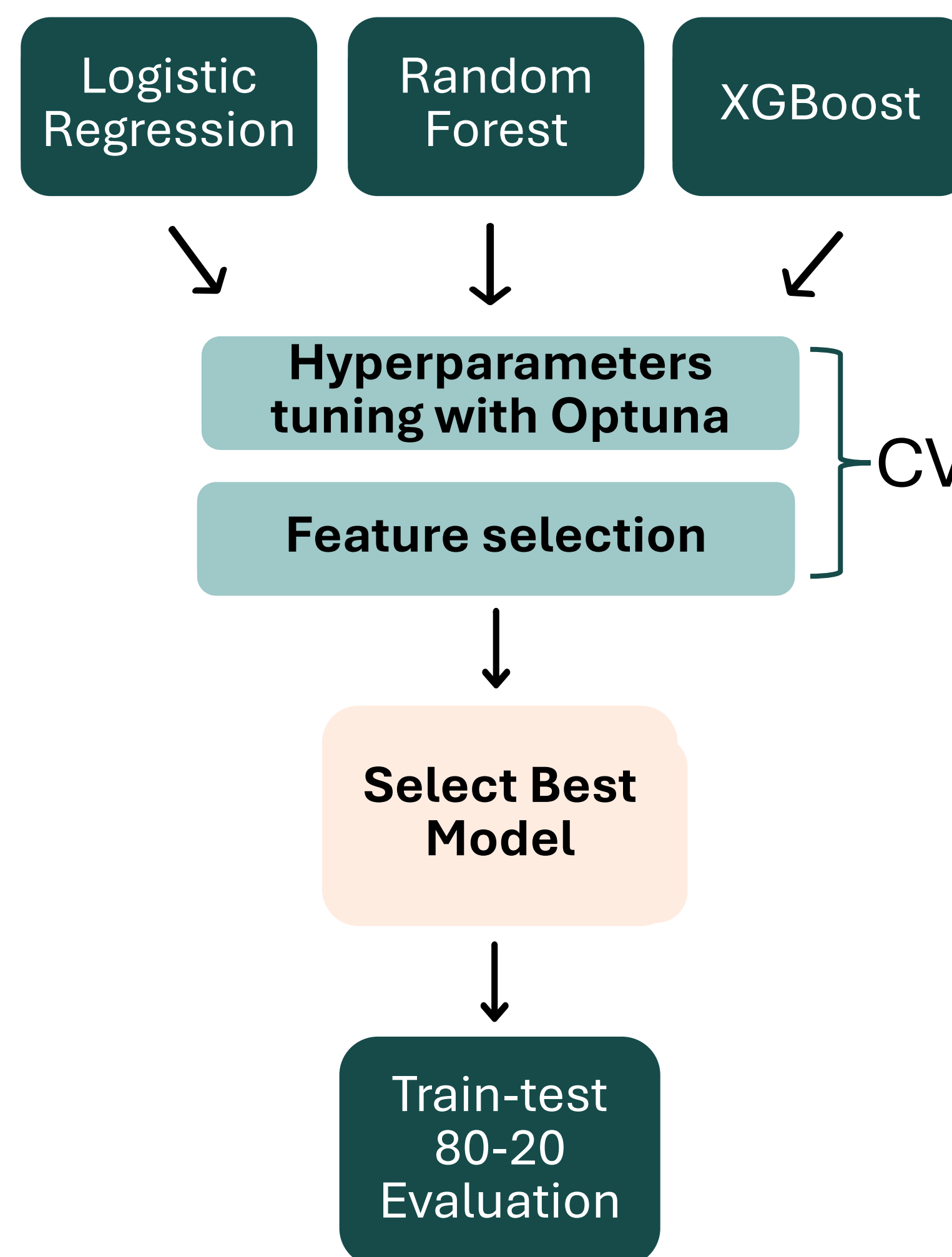
Steps:

- Sample¹ all subsequences of length 1–4 based on their frequency in the data.
- For each subsequence, calculate:
 - logits_score: aggregate² of its words' logits.
 - pos_score: aggregate² of its POS tags.
- If any subsequence has logits_score < logit_threshold³ or⁴ pos_score > pos_threshold³, classify sentence as 1 (hallucination); otherwise, classify it as 0.

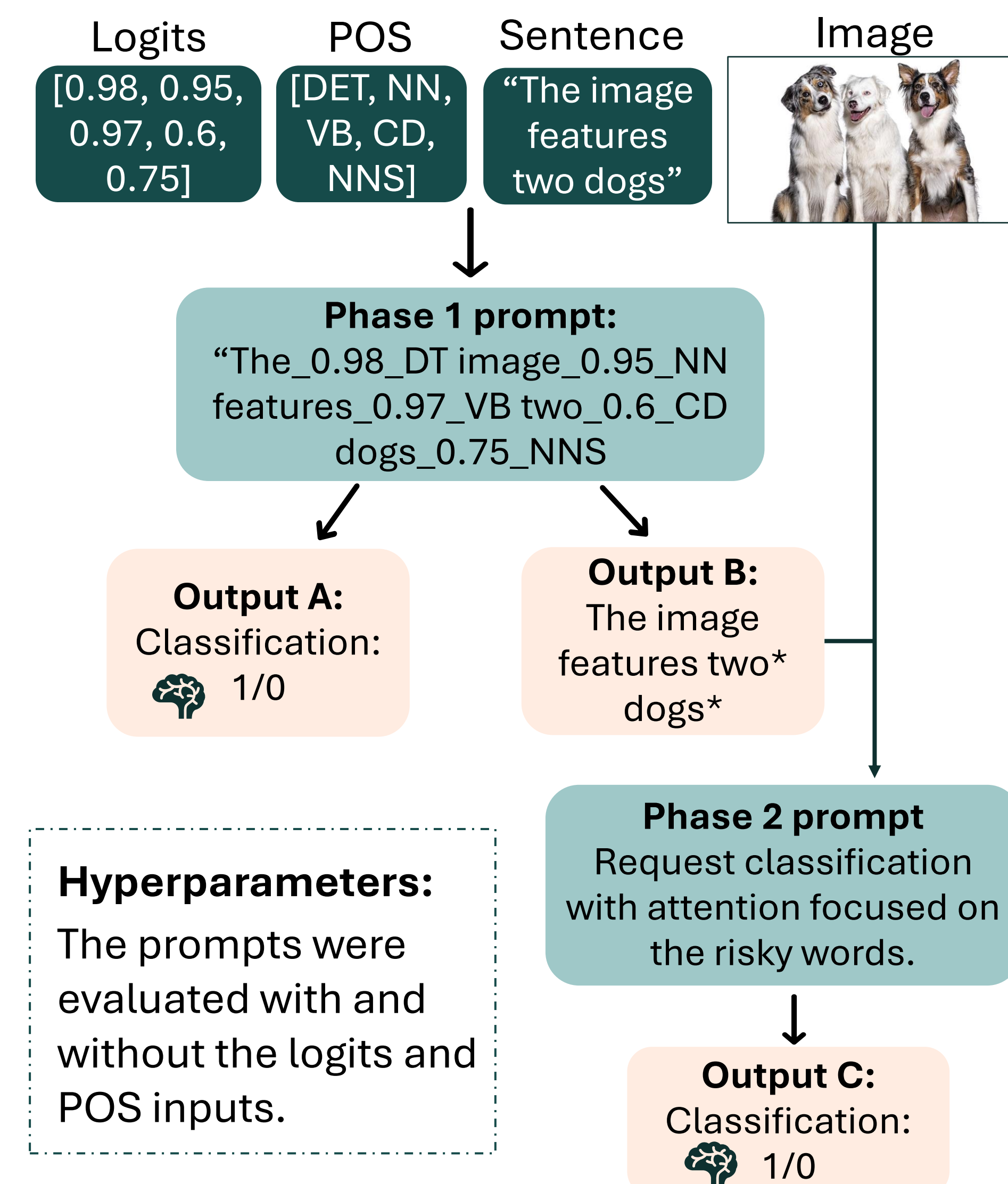
Hyperparameters:

- Sample as described or iterate all subsequences
- Aggregation functions- max, mean, median
- Different sets of thresholds for each subsequence length
- Condition: and/or/remove the POS part

Classic ML Model



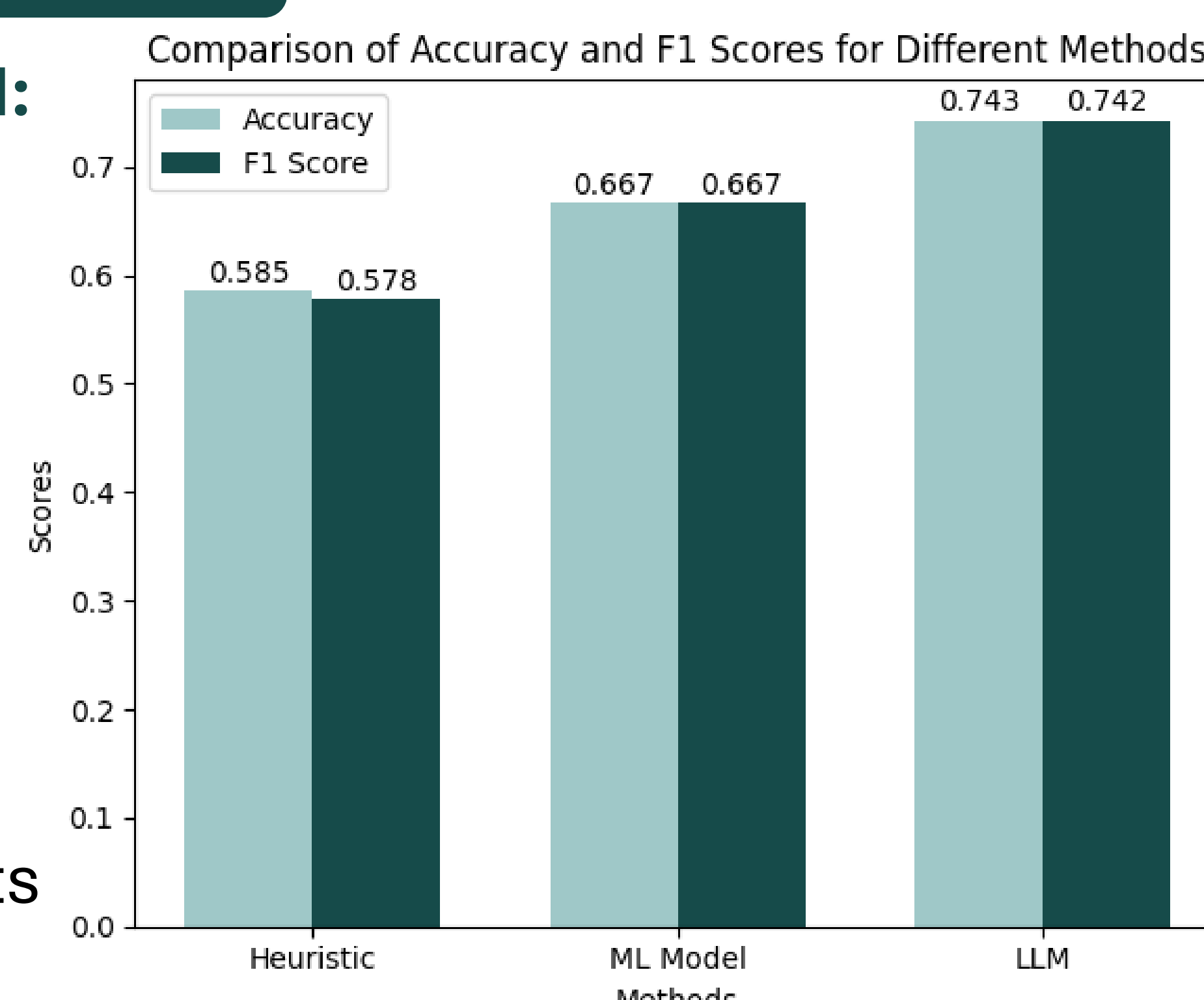
LLM



Results

Best Hyperparameters In Each Method:

- Heuristic:** sampling, mean, logits threshold: {1: 0.13, 2: 0.37, 3: 0.22, 4: 0.25}, without using POS condition.
- ML Model:** Random forest with: {'n_estimators': 244, 'max_depth': 20, 'min_samples_split': 19, 'max_features': 'log2'} and feature selection of top 10 features.
- LLM:** 2 phases prompts including logits and POS.



Conclusions

- Common hallucinations observed in data collection included incorrect object counts, colors, and relationships.
 - Hallucinations in language-vision models were effectively detected.
 - Dividing the LLM task into two phases significantly improved results (F1 score increased from 0.68 to 0.74), highlighting the benefit of simplifying complex tasks.
- Next steps:** testing methods on a larger dataset and applying the few-shot approach with LLM.